

MTH205 Analysis in one variable

Tutorial 2

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- (1) Find an open cover of $\mathbb{N}$ that has no finite subcover.
- (2) Let K be a non-empty compact subset of $\mathbb{R}$. Then show that both $\inf K$ and $\sup K$ exist and belong to K .
- (3) If $\{K_n : n \in \mathbb{N}\}$ is a sequence of non-empty compact sets in $\mathbb{R}$ such that

$$K_1 \supseteq K_2 \supseteq K_3 \supseteq \cdots K_n \supseteq \cdots,$$

then prove that $\bigcap_{n=1}^{\infty} K_n \neq \emptyset$.

- (4) Give an example of a disjoint closed subsets F_1, F_2 of $\mathbb{R}$ such that

$$0 = \inf \{|x_1 - x_2| : x_1 \in F_1, x_2 \in F_2\}.$$

- (5) Is Cantor set a nowhere dense set ? Justify your answer.
- (6) Prove the following:
 - (a) If $E \subset \mathbb{R}$ then E is nowhere dense if and only if $\mathbb{R} \setminus \bar{E}$ is dense set.
 - (b) $\mathbb{R}$ can not be written as a union of countable collection of nowhere dense sets.

Definition 0.1. Let $A \subseteq \mathbb{R}$. Then a point $x \in A$ is said to be *interior point* if there exists $\epsilon > 0$ such that

$$\mathcal{N}_\epsilon(x) = (x - \epsilon, x + \epsilon) \subseteq A.$$

A point $y \in \mathbb{R}$ is called a *cluster point* or *limit point* of A if for every $\epsilon > 0$, the set $\mathcal{N}_\epsilon(y) \setminus \{y\} \cap A$ is non-empty.

The set of all interior points of A is denoted by $\text{int}(A)$ and the collection of all limit points of A is denoted by $\bar{A}$.

Note that $\text{int}(A)$ is the largest open set contained in A . Whereas, $\bar{A}$ is the smallest closed set containing A .

Definition 0.2. A subset A of $\mathbb{R}$ is said to be

- *dense set* if $\bar{A} = \mathbb{R}$.
- *nowhere dense* if $\text{int}(\bar{A}) = \emptyset$.